The International Journal of Advanced Manufacturing Technology (2026) 144:355–366
https://doi.org/10.1007/s00170-026-17483-1

ORIGINAL ARTICLE

# Combining demand classification and forecasting models in spare parts inventory management for the energy sector

Leandro Fernandes da Silva[1] · Eliane da Silva Christo[1] · Kelly Alonso Costa[1] · Hugo Pimentel Tavares[1]

Received: 4 August 2025 / Accepted: 12 January 2026 / Published online: 28 March 2026
© The Author(s) 2026

**Abstract**

Industries are going through a changing scenario, in which they need to constantly seek to improve their processes and increase productivity. One of the factors that contribute to improving the performance of companies is inventory management, which starts with forecasting demand. This research aimed to compare the Machine Learning models, Random Forest and XGBoost, and the classic Exponential Smoothing prediction models, Holt, Holt-Winters and SARIMA Seasonal Autoregressive Integrated Moving Average. Identify which model generates the best forecasts for the items of a company in the energy sector in southeastern Brazil. The database consisted of 2,244 items. The process of classifying the demand was carried out in four types: Irregular, Intermittent, Erratic and Regular. The process of adding financial value was applied in three groups: A, B and C. The prediction models were implemented using Spyder (Python 3.11), performing 13,464 iterations. To measure the performance of the prediction models, it was, the Criteria Importance Through Intercriteria Correlation (CRITIC) multicriteria analysis method was used to determine the score of the models based on the error measures. The Augmented Dickey-Fuller tests were performed (ADF), Kwiatkowski-Phillips-Schmidt-Shin (KPSS), Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). The result of the analysis identified that, in general, the model that presented the best predictions was Random Forest, followed by ExpSmoothing. The model Random Forest presented better forecasts for items with irregular and intermittent demand, Holt-Winters performed better for regular items, while items with erratic demand did not present models with significant prominence. It also identified that the 82 items with value classification A are subject to a more detailed analysis, due to the high representativeness of the value.

**Keywords** Forecasting · Machine learning · Holt-Winters · SARIMA · Demand classification · Spare parts

## 1 Introduction

Industries are going through a changing scenario, in which they need to constantly seek to improve their processes and increase productivity to ensure the maintenance of operations. The rapid globalization of the world economy requires companies to constantly seek to increase the competitiveness of their operations. The growth of technological development allows companies to participate in global markets in the same way, making it possible for companies from other countries to participate in markets that were previously regional [1].

Companies need to adapt to deal with complex factors such as trade protectionism and growing trends in political risks. This adaptation is an urgent and vital issue for organizations [2].

✉ Leandro Fernandes da Silva
leandro.fern.silva@gmail.com

Eliane da Silva Christo
elianechristo@id.uff.br

Kelly Alonso Costa
kellyalonso@id.uff.br

Hugo Pimentel Tavares
hugot@id.uff.br

[1] Postgraduate Program in Production Engineering, Federal Fluminense University (UFF), Volta Redonda, RJ 27255-125, Brazil

Through the use of various computing technologies, companies can optimize their business decisions by using computing to analyze a large amount of available data [3].

Inventory management is one of the factors that helps improve the performance of companies, since inventory holding costs represent between 20 and 40% of their value [1].

They investigate the application of demand forecasting models for time series, more specifically the ARIMA, exponential smoothing and Holt-winters models. To predict the demand for spare items of a steel company located in the state of Rio de Janeiro, RJ, Brazil, a process of classification of the historical series was carried out and a sample of the research base of 40 historical series was used. The feasibility of using demand forecasting models based on the accuracy of the models was studied [1].

This work aims to analyze and propose the best forecasting models for each of the two types of demand classification addressed in this research, using the exponential smoothing forecasting models, Holt, holt-winters and SARIMA. Two types of classification will be carried out, one by ABC classification of the value and by classification according to the type of demand. The research was developed in an energy company located in the state of Minas Gerais, MG, Brazil, using a total base of 2,244 historical series of consumption of spare items.

Both contributed significantly to time series forecasts. They differ in their specific objectives, the volume of data analyzed, market segment, regional focus, and methodological approach. The first article emphasizes the feasibility analysis of the use of demand forecasting models, using a sample of 40 historical series, the research developed in a steel company in the state of Rio de Janeiro, while the second proposes the most suitable forecasting models based on the accuracy of the models, does not use a sample base, but the total base of 2,244 historical series. and developed the research in an energy company located in the state of Minas Gerais [1].

The novelty of this research lies in the methodology and objective: the total population of the historical series was used to process the forecasts and not a sample base, avoiding sample selection bias and generating greater precision and reliability of the results. The SARIMA model was also used to make the predictions and an ABC classification by value was made according to the Pareto principle. As a result, this study has the potential to propose more assertive models, reducing forecasting errors and indicating a specific model for each type of demand: irregular, intermittent, erratic and regular.

One of the challenges to carry out material planning is the correct sizing of inventory levels, that is, achieving the lowest possible value of fixed capital cost that meets the needs of customers, without the risk of shortage or excess inventory.

There are some factors that make it difficult to make decisions in inventory management: absence of data, presence of exceptions, dispersion in consumption history, high delivery times, delays and slowness in the purchase process, lack of information about maintenance plans, among other factors.

Based on this scenario, in which organizations need to constantly improve their performance and taking into account the high complexity of inventory planning for energy companies, this research aims to analyze and propose the best demand forecasting models according to the behavior of the historical series.

## 2 Fundamentals

### 2.1 Criteria importance through intercriteria correlation

The Criteria Importance Through Intercriteria Correlation (CRITIC) method is an analytical approach designed to determine objective weights of importance in multicriteria decision problem (MCDM) [4].

Unlike other methods that require subjective expert judgments, CRITIC draws these weights directly from the intrinsic information in the data, based on the variability of each criterion and its independence or conflict with the other criteria [5].

In addition, it avoids subjective interventions by not requiring the analyst to set arbitrary cut-off values to deal with highly correlated criteria [4].

### 2.2 Model selection measures

Error measures are statistical metrics that quantify the difference between the values projected by a model and the actual observed data [6, 7]. They serve to validate the accuracy and reliability of forecasts, allowing different methods to be compared and the most rigorous model to support decision-making [5, 6].

Mean Absolute Error (MAE) is a metric that calculates the average magnitude of the absolute differences between the values predicted by a model and the actual observed data [6–8]. It serves to validate the accuracy and performance of forecasting models, allowing you to compare different methods to select the projection that is technically more faithful to reality [6, 9, 10].

MAPE expresses the average of absolute errors in percentage terms in relation to the actual observed values [7, 8]. Measures the accuracy of forecasting models, making it

easier to compare performance between different statistical methods [6, 10, 11].

The sMAPE evaluates the accuracy of forecasts by integrating actual and projected values to balance the calculation of deviations. Compares the performance of different models, ensuring a more robust technical validation that is less biased by the scale of the data analyzed [5].

The RMSE is the square root of the mean of the quadratic deviations between the predicted values and the actual data, quantifying statistical accuracy. It serves to validate the accuracy of models, emphasizing larger errors (outliers) to ensure more technically reliable and rigorous projections [1, 2].

They explain that the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are fundamental statistical criteria for the selection and comparison of models, balancing fit to data with technical simplicity [8].

They serve to identify the best parameter configuration, penalizing excessive complexity to avoid overfitting error [11].

In practice, the aim is to minimize these values in order to find the model that explains the data with the least possible number of variables. In this way, they ensure the choice of the most efficient and statistically valid architecture to support predictions and decision-making [5].

### 2.3 Machine learning

Machine Learning is a set of techniques and algorithms that allow systems to automatically learn patterns and representations from raw data [6, 11]. It stands out for its ability to self-learn and generalize, surpassing traditional statistical models when dealing with nonlinear and complex relationships [5].

Its function is to perform tasks such as demand forecasting, item classification and pattern detection, ensuring greater accuracy in large volumes of information [7, 9, 12]. In short, it serves to transform Big Data processing into valuable insights, optimizing industrial and commercial processes, and strategic decision-making [6, 9, 13].

### 2.4 XGBoost

Extreme Gradient Boosting (XGBoost) is an advanced and efficient implementation of gradient boosting algorithms that uses decision tree-based models [7, 9].

It employs sophisticated regularization techniques to prevent overfitting and process large volumes of data in a scalable manner, ensuring high technical accuracy [9].

Its purpose is to make robust forecasts in various sectors, such as B2B commercial demand, seasonal time series, and industrial processes, surpassing traditional statistical models. Due to its fast training and low processing time, it serves as an essential tool for real-time control systems and strategic decision-making support [7].

### 2.5 Random forest

Random Forest is an ensemble learning algorithm that combines predictions from multiple decision trees to increase accuracy and technical robustness [9].

It uses the bootstrapping method to train independent trees on different subsets of data, which reduces variance and prevents overfitting. Its primary purpose is to process high-dimensional data and identify complex patterns [13].

### 2.6 Exponential smoothing

Exponential time series smoothing methods have two types, with trend and/or seasonal pattern and without them. These predictions are the weighted averages of previous observations, with weights decaying exponentially as observations age [14].

Simple exponential smoothing is one of the easiest models to apply to forecasting problems [14].

### 2.7 Holt model

Charles C. Holt adapted simple exponential smoothing to enable forecasting of trended data [15]. Holt's method assumes that datasets have a trend and do not have seasonality, so it uses two components, level and trend. This method is mainly used for linear trends with short to medium forecast periods.

### 2.8 Holt-Winters model

Holt-Winters is an extension of Holt's method developed by Holt's student Peter Winters that assumes the existence of seasonal and trend variations for data observation [16]. Considers level, trend and seasonality with the corresponding parameters alfa $(x)$, beta $(B)$ e gama $(y)$.

The Holt-Winters model is one of the most widely used methods for analyzing time series with seasonal and trend patterns [14].

### 2.9 SARIMA and ARIMA model

The Seasonal Autoregressive Integrated Moving Average (SARIMA) model can handle non-stationary time series and capture trends, seasonality, and other linear information in the data, thus enabling modeling and forecasting [17].

The seasonal autoregressive integrated moving average (SARIMA) is a model for forecasting time-series data,

including seasonal components [18]. Different experiments have concluded that SARIMA is the most suitable method for time series forecasting, estimates can be used for long-term forecasts by modifying and adding new values [19, 20].

The seasonal autoregressive integrated moving average (SARIMA) model, proposed by Box and Jenkins, has been one of the most popular approaches in seasonal time series forecasting [21].

The Autoregressive Integrated Moving Average (ARIMA) or Seasonal ARIMA (SARIMA) models were used by [22], And they commented that it is a model that considers the uncertainty of production demand and the condition of production equipment to increase the efficiency of the system and minimize maintenance costs according to specific characteristics.

Regression models are often the simplest approach and require less data pre-processing [23].

The Autoregressive Integrated Moving Average (ARIMA) model was introduced by Box and Jenkins in the 1970s. In addition, they comment that it is a more popular technique and of wide application in time series forecasting [1].

They tested fourteen demand forecasting methods for Time Series in their research and reported that the Autoregressive Integrated Mobile (ARIMA) method achieved good results in forecasting [24].

The ARIMA model has a versatility in handling various components of a time series, including autoregressive behavior (AR), differential effects (I) and moving averages (MA) being particularly suitable for data from the ARIMA [25].

They used time series forecasting methods for energy forecasting in IoT environments and big data, enabling continuous monitoring of an organization's energy consumption in real time [26]. Prediction models have one thing in common, they require a large database to achieve representative results [23].

### 2.10 Classification ABC

ABC analysis is based on the Pareto Principle, defined by the Italian economist Vilfredo Pareto, which says that the majority values (80% of its value) of a given group come from a relatively small portion of its constituents (20% of your quantity).

The fundamental inventory management technique used to categorize items according to their importance, value, and financial impact, it allows organizations to focus their resources and control efforts on the most critical items, rather than treating all products uniformly, which would be economically unfeasible [12].

The most applied method to aggregate products is ranking ABC, that determines the importance of the product [27].

### 2.11 Methodology

Regarding the methodology, this study is applied, exploratory, and quantitative in nature, as it aims to apply acquired knowledge to compare demand forecasting models for time series, ultimately contributing to improved demand forecasting performance in the energy company. In addition, a bibliographic review and database extraction were conducted.

## 3 Research development

The stages of development of this study, from the literature review to the final conclusions, are illustrated in Fig. 1.

The eight stages of the research development process can be observed in Fig. 1 The process begins with data extraction, followed by data processing, and concludes with the performance analysis of the models and identification of the best-performing model.

### 3.1 Data collection

This research was conducted in a large energy company located in the southeastern region of Brazil. The data were

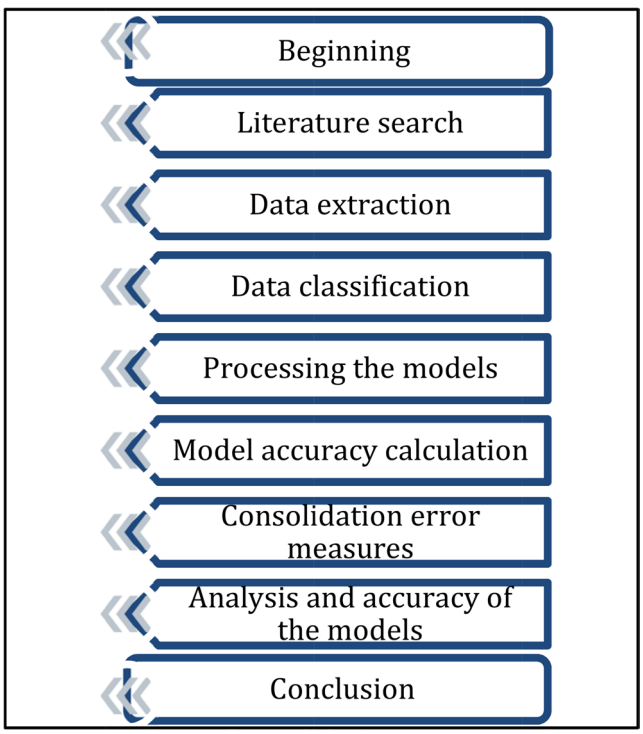



**Fig. 1** Work development flowchart. Source Authorship (2025)

extracted from the system Enterprise Resource Planning (ERP) software of the SAP R/3 materials management module.

This company has a database of 2.244 registered stockable items. The consumption history of the 60 months, starting from January/20 to December/24.

### 3.2 Data classification

Demand classification models of [28] are intensively discussed in the literature, being the two main models.

The methodology proposed by [28] It classifies the demand of items into soft, slow and sporadic categories, based on the variance partition, whose purpose was to identify the most appropriate forecasting and inventory control methods for the resulting categories.

The classification method was adapted to four categories erratic, irregular, intermittent and regular, based on the average inter-demand interval (ADI) and the square coefficient of variation of demand sizes $\left(CV^2\right)$ where the standard deviation of the item's demand divided by the average demand [29].

The average range between demands and the square coefficient of variation of the demand sizes are represented as follows Eq. 1 and Eq. 2.

$$ADI = \frac{\sum_{n=1}^{N_{pi}} t_i^n}{N_{pi}} \tag{1}$$

$$CV_i = \frac{\frac{\sum_{n=1}^{N_{pi}} (d_i^n - d_i)^2}{N_{pi}}}{\frac{\sum_{n=1}^{N_{pi}} d_i^n}{N_{pi}}} \tag{2}$$

Consider $t_i^n$ period interval without demand for the item $i$, $N_{pi}$ number of non-zero demands in the time interval $d_i^n$, Item demand value $i$ in the period $n$ and $d_i$ average of the item's non-null demands $i$. The cutoff points for this classification $ADI = 1{,}32$ e $CV^2 = 0{,}49$..

This classification shows four regions, in which regular items have high turnover and low variation, the intermittent classification presents several periods without demand, low turnover and low randomness in the size of demand, the erratic classification presents a high turnover with high variation in the size of demand, while irregular materials are of low turnover with high variation in demand.

During the development of this work, a classification process was conducted and the items were segmented into four categories: Erratic, Irregular, Intermittent and Regular. The premise for classification was the average interval between the demands and the coefficient of the variation of the sizes of the demands. The cutoff points for classification $ADI = 1{,}32$ and $CV^2 = 0{,}49$. The Fig. 2 presents the result of the classification process.

### 3.3 Classification ABC

Subsequently, the 2,244 items were aggregated using ABC ranking. Which determined the importance of items by consumption value. The materials were grouped into three categories, as shown in the Fig. 3.

We can observe in Fig. 3 that the aggregation by category A represents most of the financial value, 80% of the consumption value, which corresponds to 82 items 4% of the materials. It may be the manager's decision in this case to make a more detailed analysis of the forecasts. Based on the information in Fig. 2, we observed that more than half of the items analyzed have a demand classified as erratic, which makes the demand forecasting process difficult, since these items with this classification have a great variability in demand.

This grouping enables decision-makers to direct their efforts, if necessary, on the most financially important items.

Since the ABC classification makes it possible to aggregate the highest consumption values for smaller quantities of items.

Figure 3 shows that items B represents 15% of the consumption value and 213 items corresponding to 10% of the products, this category concentrates a less significant part of the consumption value. The items C represent 5% of the consumption value and 1.949 Items 87% of materials.

### 3.4 Comparison of classification models

Comparing the two classification models used in this research, classification by behavior of the historical series and classification by ABC value, we can see that the two

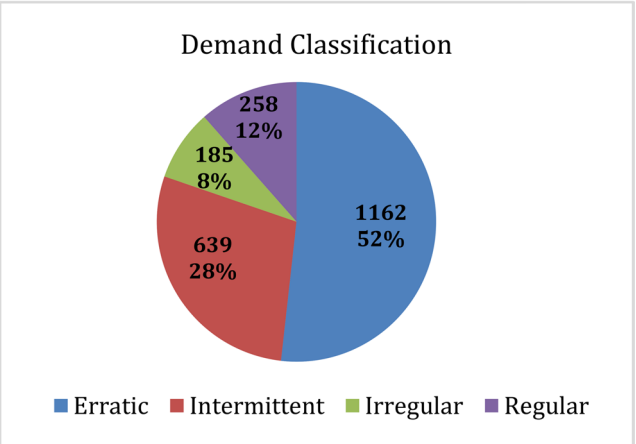



**Fig. 2** Demand Classification fo items. Source Authorship (2025)

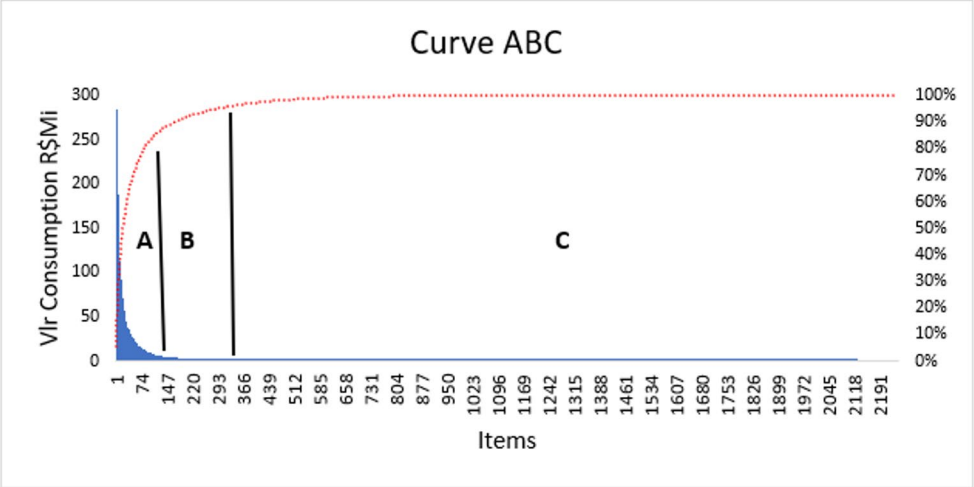



**Fig. 3** Aggregation of materials by categories ABC. Source Authorship (2025)

classification models can contribute to the optimization of demand forecasts. Although they have different approaches.

The Classification by behavior of the historical series addresses the issue of dispersion and variability of the series.

Enabling a grouping of the items into four groups, irregular, erratic, intermittent and regular. This grouping makes it possible to identify a model that generates more assertive forecasts and that is more suitable for use in each group.

The ABC grouping aims to classify the items according to the representativeness of value. This approach enables decision-makers to perform analysis more assertively, in those with the greatest financial impact. Another relevant point is the reduction of operational workload.

Because directing the analyses to more critical items eliminates many deals with low-value materials, which is usually the majority in the database.

They used the ABC Classification in their research. They comment that among all the products in category A, the first two products in the ranking, together represent 27% of the company's sales in the period considered and were chosen for analysis, with a focus on improving the quality of demand forecasts [27]. Table 1 shows a comparison between this work and what was developed by [27].

Table 1 shows that the studies have different approaches about the sector of activity of the companies that participated in the research, software used, and the form of methodology.

**Table 1** Survey comparison

| Author | Sector | Materials Feature | Demand Classification Type |
|---|---|---|---|
| Barbosa et al.,. (2015) | Alimentary | Perishable products | Classification of value aggregation ABC by type of demand behavior |
| Author (2025) | Energetic | Spare materials | ABC Value Adding Classification |

Source Authorship (2025).

But both used the ABC Classification and identified that the ABC grouping can add value to demand forecasts.

Considering the functionality and the different forms of contribution of the two classification models, it is concluded that the joint use of these two forms of classification can contribute to improve the quality of demand forecasts.

### 3.5 Demand forecasting

The objective of this research was to compare demand forecasting models for spare parts; the primary evaluation metric was forecasting accuracy. For this reason, the forecasted values themselves were not presented in detail; instead, the step-by-step execution of the forecasting procedures was reported to enable replication and future studies by other researchers.

The model comparison process was carried out for the entire dataset of 2,244 items. Using Spyder (Python 3.11), demand forecasts were generated for each item employing the six forecasting models addressed in this research: two machine learning models (Random Forest and XGBoost) and four traditional models (Exponential Smoothing, Holt, Holt–Winters, and SARIMA).

Subsequently, forecast errors were calculated to determine the most robust models.

In total, 13,464 iterations were performed in Spyder (Python 3.11), corresponding to six forecasts for each of the 2,244 items analyzed.

The historical series was divided into training bases (Jan/2020 to Dec/2023), test base (Jan/2024 to Dec/2024) and the forecast period was set at 12 steps ahead.

He characteristics of the behavior of the historical series were analyzed: seasonality, trend and autocorrelation. The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) tests were performed, and the

Augmented Dickey-Fuller (ADF), Kwiatkowski-Phillips-Schmidt-Shin (KPSS) results were performed.

The four error measures used wereMean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Symmetric MAPE (sMAPE), Root Mean Squared Error (RMSE).

Next, the score of each prediction model was calculated based on the measures of prediction error using the Criteria Importance Through Intercriteria Correlation (CRITIC) method. The selection of the best model was made by comparing the Scores of the models, and the model chosen was the one with the highest score.

## 4 Results

From data processing, analysis of the behavior of the historical series, generation of forecasts, calculations of errors of the predicted and actual value, definition of the Score of the forecast models.

The behavior of the historical series of the Sku code 386,227 was shown in Fig. 4. This same analysis was performed for all items in the database.

Figure 4 shows the behavior of the historical series in the period of the 60 months analyzed, relating the demand in relation to the historical period, separating the training periods from the test periods of the models. Next, the seasonal behavior was analyzed Fig. 5.

We observe the decomposition of the historical series and the presentation of the seasonality of the historical series in Fig. 5.

The next step was to consolidate the results of the Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests of the historical series of the total base of the 2.244 items to analyze stationarity, as shown in Table 2.

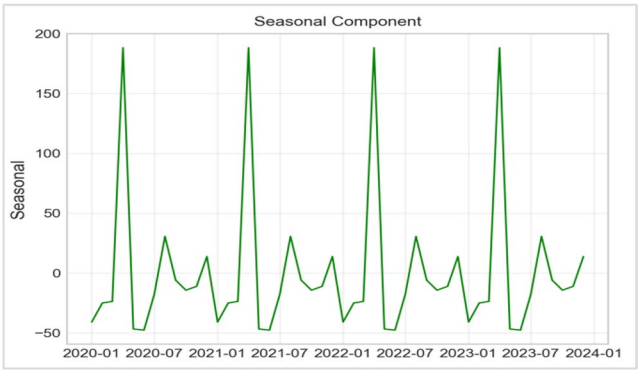



**Fig. 5** Seasonal code behavior386227. Source Authorship (2025)

Table 2 shows how2.244 time series behaved in relation to two classical stationarity tests; 61,8% of the codes have series considered stationary by both tests, presenting a high reliability, 19,7% of the items did not pass either of the two tests, possibly must be non-stationary series, 9,4% were indicated as stationary only by the ADF method and the KPSS method alone indicated that 210 as stationary.

Two other tests performed were the Akaike Information Criterion (AIC) and the Bayesian Information Criterion

**Table 2** Stationarity test resulte

| Test type | Qtd sku´s | Percentage |
|---|---|---|
| Both Tests | 1.386 | 61,8% |
| Neither | 442 | 19,7% |
| KPSS Only | 210 | 9,4% |
| ADF Only | 206 | 9,2% |
| **Total** | **2.244** | |

Source: Authorship (2025)

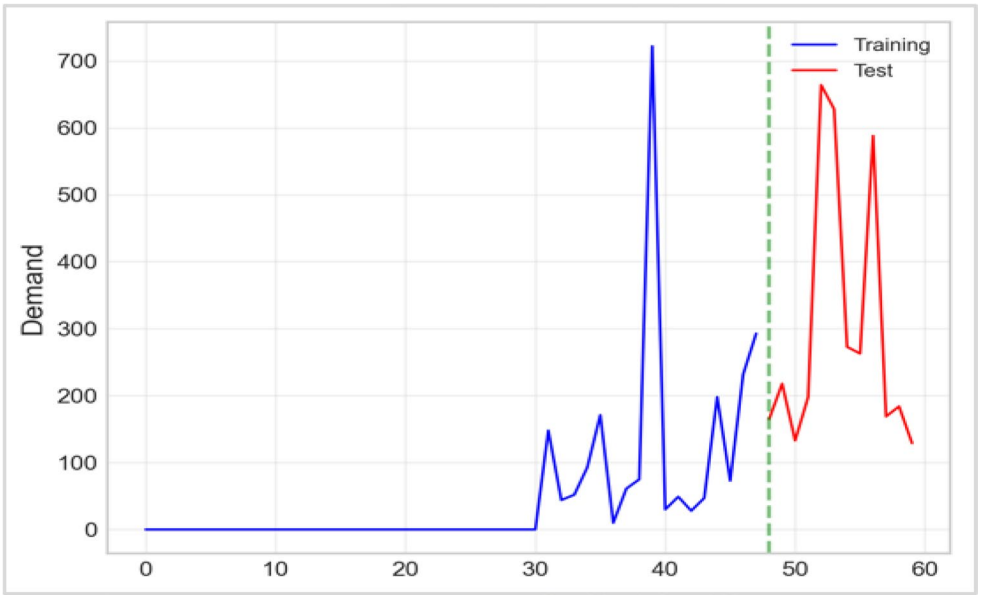



**Fig. 4** Code historical series behavior386227. Source Authorship (2025)

(BIC), Fig. 6 shows the comparison of the values of the two theses.

We identified in Fig. 6, that the points are diagonally-aligned, indicating a strong correlation, the strong correlation suggests that models with good AIC also tend to have good BIC, which facilitates the choice. The consistency between AIC and BIC reinforces the robustness of the model selection. We have identified an extreme outlier with AIC around −750, which may indicate an extremely good model or a possible miscalculation.

Based on the calculation of the score of the prediction models, it was possible to compare the models as shown in Table 3.

Based on Table 3, the model that presented the best performance score was RandomForest, which generated better forecasts for 691 SKUs out of a total of 2.244, that is 30,79% of the analyzed base.

Followed by the exponential smoothing model, predictions with greater assertiveness in relation to the other models, as it presented for 26,52% of the items in the database.

The SARIMA model appeared in the third position in performance compared to the other models. Generated forecasts with fewer errors for 294 total items and 2.044 items analyzed, that is, 13.10% of the forecasts with the best accuracy.

The Holt model and holt-winters had lower performance compared to the other models. Presenting only 10,65 and 10,16% of the forecasts with the least error. One factor that contributed to this low performance is that this model was parameterized to make forecasts that gave greater weight to consumption trends, but the largest volume in the time series does not show this type of behavior.

Table 2 shows that the Hot Winter model performed better than the Holt model. This model generated a lower error value for 22% of the predictions. This is due to the fact that the model's damping contributes to generating more assertive predictions.

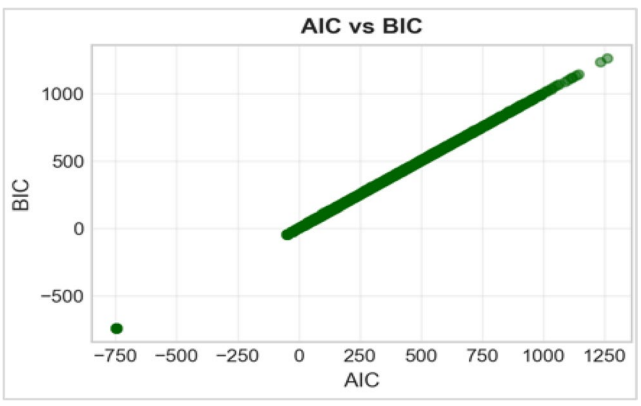



**Fig. 6** Comparison of AIC and BIC of models. Source Authorship (2025)

**Table 3** Prediction models with the highest score

| Modelo | Quantidade_SKUs | Percentual |
|---|---|---|
| RandomForest | 691 | 30,79% |
| ExpSmoothing | 595 | 26,52% |
| SARIMA | 294 | 13,10% |
| HoltWinters | 239 | 10,65% |
| Holt | 228 | 10,16% |
| XGBoost | 197 | 8,78% |
| **Total** | **2.244** | |

Source: Authorship (2025)

Based on Table 2, it is observed that the XGBoost model was the model that presented the worst performance in relation to the other models, presenting better predictions for only 8.78% of the codes analyzed.

The comparative analysis between classic forecasting models and machine learning models provides important information for the process of choosing the demand forecasting model for items in the energy sector. A model choice that provides forecasts with greater accuracy contributes to better levels of customer service and more balanced inventory levels.

Considering a total of 2.044 items in the survey, and that 12.264 iterations were made to be able to generate all the error measures of the six models surveyed, we can say that we have a very significant population to generate reliable results.

The MAE, MAPE, MASE, and RMSE error measures were calculated for each of the 2.244 historical series, and it was analyzed which model had the lowest error in the predictions. Figure 7. It shows a consolidation of the result of this analysis.

Figure 7 shows a comparison of the frequency with which the prediction models had fewer errors in the predictions, that is, each metric shows how many items each model was able to predict with the least error.

RandomForest stood out by having higher amounts of predictions with lower MAE and RMSE, which makes it ideal for applications where minimizing absolute error and large deviations.

ExpSmoothing, on the other hand, presented the lowest MAPE for forecasting 586 series, followed by the SARIMA model, which presented better forecasts for 476 historical series. The SARIMA model stood out for generating a higher volume of predictions with smaller MASE. Being useful in scenarios with large percentage variance between data.

Classic models, such as Holt and HoltWinters, have lower performance, which indicates that more modern or hybrid models may be more effective.

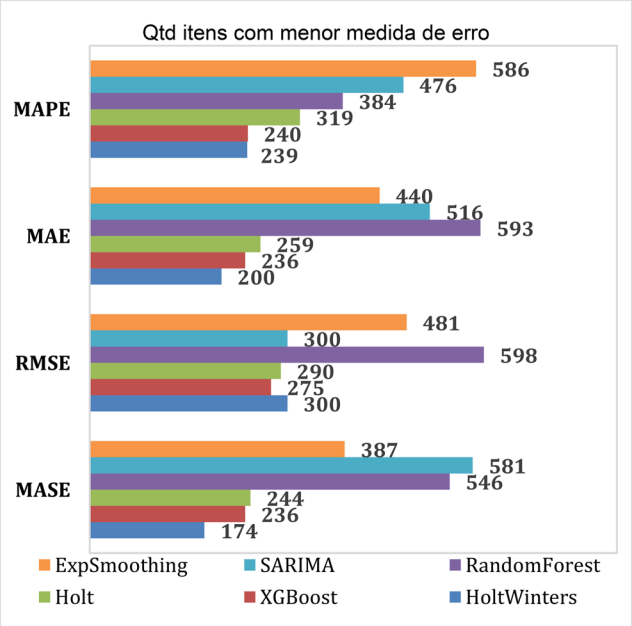



**Fig. 7** Number of times the prediction models performed best by type of error measure. Source Authorship (2025)

We can observe that it does not provide conclusive information about the effectiveness of the models, as some models stand out in one of the measures, but not in all.

One way to choose demand forecasting models can be according to the behavior of the historical series. Each historical series usually presents a different behavior, but it is possible to group them according to the characteristics of the data dispersion and the average intervals between demands. The performance of the models, according to the demand classification, is shown in Fig. 4.

Figure 8 shows that the Random Forest prediction model performed better than the other models for predicting items with intermittent, erratic, and irregular type classification.

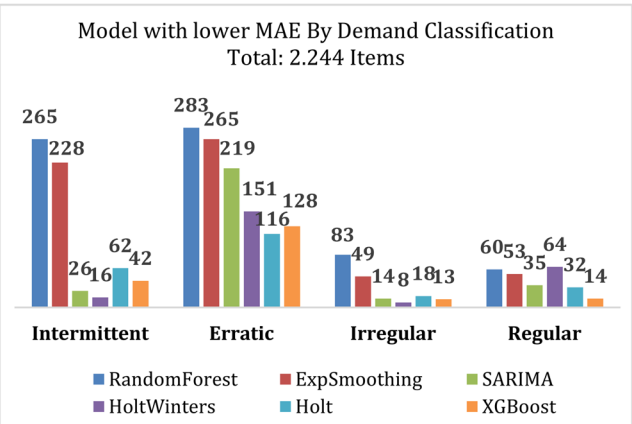



**Fig. 8** Forecasting models that showed the highest score by type of demand. Source Authorship (2025)

The EspSmoothing prediction model was the second best model with predictions for intermittent and erratic type demands. In the case of items with regular demand, the best model was Holt-Winters, which presented better forecasts, standing out from the other forecast models.

Different perspectives of analysis contribute to more accurate inferences, so Table 4 also shows the classification of the percentage of models by type of demand classification.

Based on Table 4, it can be observed that the Random Forest model presented 41% of the best predictions for intermittent items and 45% for irregular items. These percentages indicate that, as a general rule, it can be used for items with this type of demand behavior. For items with regular behavior, Holt-Winters is the most indicated, as it generated better predictions in 25% of cases.

In the case of erratic items, the Exp forecast models Smoothing and Random Forest perform similarly, and there is no model that stands out significantly.

A deeper analysis of the items of greater financial value is important in the case of materials that can have a great impact on the cost of fixed capital. Comparison of the performance of the models by ABC aggregation can be useful in these cases and has been demonstrated in Fig. 9.

Based on Fig. 9 we can see that we have only 82 items with classification A, which represents 80% of the value, and the model that presented the best result for this type of grouping was Holt-Winters. It is up to decision makers to opt for a more in-depth analysis in these cases, as they are the most financially relevant items and more assertive forecasts can generate financial gains for organizations.

However, it did not identify an ideal model that would generate forecasts with smaller errors for all types of historical series; In some cases, certain models are less wrong than others. Therefore, it is important to use model comparisons to evaluate the demand forecast, whenever there is new data and new scenarios.

The complexity of inventory planning, in view of factors such as the dispersion in the consumption history, the lack of information on maintenance plans and the presence of exceptions in the data, was evidenced by the fact that 52% of the 2.244 items analyzed have a demand classified as irregular, which significantly hinders the assertiveness of traditional projections.

To mitigate these technical risks, the methodology employs a classification based on demand behavior (ADI and CV) and financial representativeness (ABC Curve), allowing the identification of Class A items, which represent 80% of the consumption value and require much more rigorous monitoring and analysis.

The application of Machine Learning models, such as Random Forest, has proven to be technically superior to deal with this variability in intermittent and irregular demands,

**Table 4** Ranking of models by demand classification

| | Holt Winters | Holt | SARIMA | Exp Smoothing | Random Forest | XG Boost | Total |
|---|---|---|---|---|---|---|---|
| Intermittent | 16 | 62 | 26 | 228 | 265 | 42 | 639 |
| % | 3% | 10% | 4% | 36% | 41% | 7% | |
| **Tipo** | **Holt Winters** | **Holt** | **SARIMA** | **Exp Smoothing** | **Random Forest** | **XG Boost** | **Total** |
| Erratic | 151 | 116 | 219 | 265 | 283 | 128 | 1162 |
| % | 13% | 10% | 19% | 23% | 24% | 11% | |
| **Tipo** | **Holt Winters** | **Holt** | **SARIMA** | **Exp Smoothing** | **Random Forest** | **XG Boost** | **Total** |
| Irregular | 8 | 18 | 14 | 49 | 83 | 13 | 185 |
| % | 4% | 10% | 8% | 26% | 45% | 7% | 100% |
| **Tipo** | **Holt Winters** | **Holt** | **SARIMA** | **Exp Smoothing** | **Random Forest** | **XG Boost** | **Total** |
| Regular | 64 | 32 | 35 | 53 | 60 | 14 | 258 |
| % | 25% | 12% | 14% | 21% | 23% | 5% | |
| **Total** | **239** | **228** | **294** | **595** | **691** | **197** | **2244** |

Source: Authorship (2025)

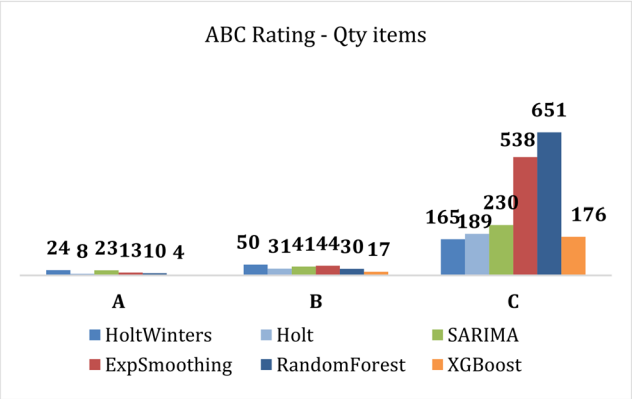



**Fig. 9** Models with lower Score by classification ABC. Source Authorship (2025)

while the use of statistical criteria such as AIC and BIC ensures the choice of the most efficient model architecture, avoiding excessive complexity and overfitting error.

It is recommended that companies adopt the practice of classifying their inventories simultaneously by the behavior of demand (ADI and CV) and by financial importance (ABC Curve). This approach allows you to target efforts surgically, the ABC curve prioritizes where accuracy is financially vital.

Instead of using a single template for all inventory, it is recommended to use Random Forest for items with irregular and intermittent demand. For items of regular and seasonal demand, the use of the Holt-Winters model should be expanded, as it has demonstrated greater technical adherence in these cases.

Detailed and timely analysis practices must be mandatorily applied to Class A items. Since only 82 items (4% of the total) represent 80% of the consumption value, any forecast error in these SKUs has a disproportionate impact on the cost of capital fixed in inventory.

Technical sectors should replace subjective model choices with multicriteria analysis methods such as CRITIC. This ensures that the selection of the prediction model is based on the variability and intrinsic conflict of metrics such as MAE, MAPE, sMAPE, and RMSE resulting in a statistically robust choice.

Sectoral practices must evolve from the use of small samples to the processing of the total population of data (Big Data). The use of tools such as Python allows thousands of iterations (such as the 13.464 mentioned), eliminating selection biases and increasing strategic reliability.

As a limitation, it is noteworthy that the application of the survey results in inventory management requires the performance of additional calculations related to specific parameters of inventory planning, with safety stock, order point. For future studies, it is suggested to explore the use of forecasts as a support for inventory sizing and management.

And finally, it is recommended to constantly reevaluate the models whenever new scenarios or data emerge, since no model is ideal for all types of historical series permanently.

## 5 Conclusions

It was possible to achieve the objective of this work, which was to compare Machine learning models and classical models for forecasting demand for spare items of a company in the energy sector. The Machine Learning models used were Random Forest, XGBoost, and the classic models Exponential Smoothing, Holt, Holt-Winters, and SARIMA.

To achieve this objective, it was necessary to establish two types of classification. Classification of demand for

items as erratic, irregular, intermittent and regular. The result of the classification process showed that 52% of the items have irregular demand, which makes the demand forecasting process difficult.

Another classification process used was value aggregation, in which the items were grouped into ABC categories.

In total, the demand forecast for 2,244 spare items was analyzed. In general, the model that presented the score in the demand forecasts was Random Forest and the second best model was ExpSmoothing.

Analyzing the accuracy of the models based on the demand classification, it was identified that the most appropriate model for items with irregular and intermittent demand was the Random Forest; for regular items, the most adherent model was Holt-Winters, while for items with erratic demand there was no model with significant prominence.

It concludes that Random Forest is superior for irregular and intermittent demands, in line with similar research [7, 9, 11], that validate the primacy of ensemble and deep learning methods in complex scenarios.

This research confirms that the traditional SARIMA and Holt-Winters models have limited utility to regular series and stable seasonalities, corroborating the findings of other researchers [10].

The classification approach by grouping of values showed that, of the total items analyzed, only 82 items were grouped in category A, with greater representativeness of value. And for these items, a more punctual analysis is recommended, as they are items that can cause a great financial impact on the cost of fixed capital.

Finally, it was possible to meet the objective of this work by comparing the demand forecasting models based on the extracted data and on the measures of accuracy of the forecasts and to identify the best model.

**Author contributions** Conceptualization, L.F.S., E.S.C., and K.L.C.,; data curation, L.F.S., E.S.C., H.P.T., and K.L.C.,; formal analysis, L.F.S., E.S.C., H.P.T., and K.L.C.,; funding acquisition, L.F.S., E.S.C., H.P.T., and K.L.C.,; methodology, L.F.S., E.S.C., and K.L.C.,; project administration, L.F.S., E.S.C., and K.L.C.,; resources, L.F.S., E.S.C., and K.L.C.,; software, L.F.S., E.S.C., H.P.T., and K.L.C.,; supervision, L.F.S., E.S.C., and K.L.C.,; validation, L.F.S., E.S.C., and K.L.C.,; writing—original draft, L.F.S., E.S.C., and K.L.C.,; writing—review and editing, L.F.S., E.S.C., and K.L.C.,. All authors have read and agreed to the published version of the manuscript.

**Funding** The Article Processing Charge (APC) for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614). The authors acknowledge the support of the Production Engineering Program at Fluminense Federal University (PPGEP/UFF). The Brazilian funding agencies CNPq -Conselho Nacional de Desenvolvimento Científico e Tecnológico and CAPES - Coordenação de Aperfeiçoamento de Pessoal de Nível Superior. Research group Decision Support Systems.

**Data availability** The data that support the findings of this study are available from the corresponding author upon reasonable request.

## Declarations

**Institutional review board statement** Not applicable.

**Informed consent** Not applicable.

**Conflict of interest** The authors declare no conflict of interest.

**Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

## 5 References

1. Da LF et al (2024) Silva Demand Forecasting for Steel Company Spare Items with Time Series Templates, *Procedia Computer Science*, Elsevier B.V., pp. 57–64. https://doi.org/10.1016/j.procs.2024.08.230
2. Sui X, Jiao S, Wang Y, Wang H (2024) Digital transformation and manufacturing company competitiveness. Financ Res Lett 59:104683. https://doi.org/10.1016/J.FRL.2023.104683
3. Wang F, Aviles J (2023) Enhancing operational efficiency: integrating machine learning predictive capabilities in business intellgence for informed decision-making. Frontiers in Business, Economics and Management 9(1):282–286. https://doi.org/10.54097/FBEM.V9I1.8694
4. Diakoulaki D, Mavrotas G, Papayannakis eL (1995) Determining objective weights in multiple criteria problems: the critic method
5. Tavares HP et al (2025) Complementarity of renewable energy sources in Brazil using qualitative comparative analysis of causal configurations for SDG 7.2 achievement. Discover Sustain. https://doi.org/10.1007/s43621-025-02480-3
6. Peng ZJ, Zhang C, Tian E (2023) Crude oil price time series forecasting: a novel approach based on variational mode decomposition, time-series imaging, and deep learning. IEEE Access 11:82216–82231. https://doi.org/10.1109/ACCESS.2023.3301576
7. Kramar V, Alchakov V (2023) Time-series forecasting of seasonal data using machine learning methods. Algorithms. https://doi.org/10.3390/a16050248
8. Boukrouh I, Idiri S, Tayalati F, Azmani A, Bouhsaien L (2025) A hybrid approach for sales forecasting: combining deep learning and time series analysis. Int J Eng 38(4):859–870. https://doi.org/10.5829/ije.2025.38.04a.16
9. Lim PY, Chin WY, Kong and YH, Exploring the Effectiveness of Ensemble Learning for Business-to-Business (B2B) Demand Forecasting: An Empirical Analysis, em (2024) - *Conference Proceedings: 2024 3rd International Conference on Digital* Transformation and Applications, Institute of Electrical and Electronics

Engineers Inc., 2024, pp. 226–230. https://doi.org/10.1109/ICDXA61007.2024.10470870
10. Munim ZH, Fiskin CS, Nepal B, Chowdhury MMH (2023) Forecasting container throughput of major Asian ports using the Prophet and hybrid time series models. The Asian Journal of Shipping and Logistics 39(2):67–77. https://doi.org/10.1016/j.ajsl.2023.02.004
11. Cao Z, Han X, Lyons W, and O'Rourke F (2021) Energy management optimisation using a combined Long Short-Term Memory recurrent neural network – Particle Swarm Optimisation model, *J Clean Prod*, vol. 326, dez. https://doi.org/10.1016/j.jclepro.2021.129246
12. Abdolazimi O, Shishebori D, Goodarzian F, Ghasemi P, Appolloni A (2021) Designing a new mathematical model based on ABC analysis for inventory control problem: a real case study. RAIRO - Operations Research 55(4):2309–2335. https://doi.org/10.1051/ro/2021104
13. Costa VBF, Scianni L, Miranda RC, Bonatto B (2023) Assessment of the status and trends of photovoltaic distributed generation in Brazil: an in-depth approach based on big data processing. Solar Energy 249:694–711. https://doi.org/10.1016/j.solener.2022.12.018
14. Kumar V, Unal S, Bhagat SK, Tiyasha T (2024) A data-driven approach to river discharge forecasting in the Himalayan region: insights from Aglar and Paligaad rivers. Results Eng 22:102044. https://doi.org/10.1016/J.RINENG.2024.102044
15. Holt CC (2004) Forecasting seasonals and trends by exponentially weighted moving averages. Int J Forecast 20(1):5–10. https://doi.org/10.1016/J.IJFORECAST.2003.09.015
16. Winters PR Forecasting sales by exponentially weighted moving averages on JSTOR. Accessed on May 5, 2025. [Online]. Available at: https://www.jstor.org/stable/2627346
17. Lu L, Yang T, Chen Z, Ge Q, Yang J, Sen G (2025) Prediction analysis of human brucellosis cases in Ili Kazakh Autonomous Prefecture Xinjiang China based on time series. Sci Rep 15(1):1–13. https://doi.org/10.1038/S41598-024-80513-Z
18. Schuh G, Schultze W, Schiffer M, Rieger A, Rudolf S, Lehbrink H (2014) Scenario-based determination of product feature uncertainties for robust product architectures. Prod Eng Res Devel 8(3):383–395. https://doi.org/10.1007/S11740-014-0532-4/FIGURES/11
19. Shahin M, Chen FF, Maghanaki M, Firouzranjbar S, Hosseinzadeh A (2025) Evaluating the fidelity of statistical forecasting and predictive intelligence by utilizing a stochastic dataset. Int J Adv Manuf Technol 138(1):193–223. https://doi.org/10.1007/S00170-024-14505-8/METRICS
20. Shahin M, Maghanaki M, Hosseinzadeh A, Chen FF (2024) Improving operations through a lean AI paradigm: a view to an AI-aided lean manufacturing via versatile convolutional neural network. The International Journal of Advanced Manufacturing Technology 2024 133:11 133(11):5343–5419. https://doi.org/10.1007/S00170-024-13874-4
21. Pai PF, Lin CS (2005) Using support vector machines to forecast the production values of the machinery industry in Taiwan. Int J Adv Manuf Technol 27(1–2):205–210. https://doi.org/10.1007/S00170-004-2139-Y/METRICS
22. Bahou Z, Lemnaouar MR, Krimi I (2024) Integrated non-cyclical preventive maintenance scheduling and production planning for multi-parallel component production systems with interdependencies-induced degradation. Int J Adv Manuf Technol 130(9–10):4723–4749. https://doi.org/10.1007/S00170-024-12975-4/METRICS
23. Martin M, Gneiting S, Benfer M, Lanza G (2024) Incentive system to smooth out fluctuations in demand. Prod Eng 19(1):157–171. https://doi.org/10.1007/S11740-024-01300-3/FIGURES/5
24. Effrosynidis D, Spiliotis E, Sylaios G, Arampatzis A (2023) Time series and regression methods for univariate environmental forecasting: an empirical evaluation. Sci Total Environ 875:162580. https://doi.org/10.1016/J.SCITOTENV.2023.162580
25. Bindel LJ, Seifert eR Long-term forecast for antibacterial drug consumption in Germany using ARIMA models. Naunyn Schmiedebergs Arch Pharmacol 1–20. https://doi.org/10.1007/S00210-024-03721-4/TABLES/9
26. Kim M, Lee S, Jeong T (2023) Time series prediction methodology and ensemble model using real-world data. Electronics 12(13):2811. https://doi.org/10.3390/ELECTRONICS12132811
27. Barbosa N, Christo E da S, Costa KA (2015) Demand forecasting for production planning in a food company, 10, 16, [Online]. Disponível em: www.arpnjournals.com
28. Williams TM (1984) Stock control with sporadic and slow-moving demand. J Oper Res Soc 35(10):939. https://doi.org/10.2307/2582137
29. Syntetos AA, Boylan JE (2001) On the bias of intermittent demand estimates. Int J Prod Econ 71(1–3):457–466. https://doi.org/10.1016/S0925-5273(00)00143-2

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.